

CSCE 671 HUMAN-COMPUTER INTERACTION · MODULE 08 · LECTURE NOTES

Controlled Experiments with People

Measuring a difference, which needs a design.

MODULE OUTCOMES

Choose between within-subject and between-subject designs.

Explain counterbalancing and why it is necessary.

Explain the statistics for small human samples.

Explain the specific threats to validity here.

Design and analyse an experiment with people.

1 Two designs

	Within-subject	Between-subject
Each person	Uses both designs.	Uses only one design.
Controls for	Individual differences — completely, since each person is their own control.	Nothing directly; it relies on randomisation to balance the groups.
Participants needed	Far fewer.	Many more — see the callout for why.
The problem	Order and learning effects (§2).	Variance between people swamps the effect you are looking for.
Use when	The task can reasonably be repeated.	Exposure to one design permanently changes how the other is approached (§2's carryover).

INDIVIDUAL DIFFERENCES ARE LARGER THAN DESIGN DIFFERENCES, WHICH IS THE GOVERNING FACT

The fastest and the slowest participant on the same task with the same interface frequently differ by a factor of several — and the difference between two reasonable designs is usually very much smaller than that.

So a between-subject design has to detect a small effect against a large variance, which requires many participants — frequently more than you can realistically recruit, which means the study is underpowered and will find nothing whichever design is better.

While a within-subject design removes that variance entirely, because each person is compared against themselves — which is why it can work with a dozen people where a between-subject design would need a hundred.

And it introduces order effects instead, which are manageable by counterbalancing (§2) — a far better trade than needing ten times the participants, and is why within-subject should be the default whenever the task can be repeated.

2 Counterbalancing

THE PROBLEM

learning	the second design benefits from the practice gained on the first
fatigue	the second design suffers from tiredness and boredom
carryover	the first design's mental model is applied to the second

SO A -> B ORDER CONFOUNDS THE DESIGN WITH THE ORDER.

THE FIX: COUNTERBALANCING

half the participants see A then B, and half see B then A. The order effect then applies equally to both conditions and cancels out in the comparison.

AND WITH MORE THAN TWO CONDITIONS

a Latin square, so that each condition appears in each position equally often.

AND CARRYOVER IS THE ONE COUNTERBALANCING DOES NOT FIX -- if using A permanently changes how somebody thinks about B, go between-subject instead.

Carryover is the one counterbalancing does not fix, and it is the condition that forces a between-subject design — so distinguishing carryover from learning determines the design choice and is worth thinking about before recruiting. Learning is symmetric and cancels; carryover is a permanent change to the participant and does not. A new conceptual model (Module 04) is the usual source of carryover: once somebody understands design A's model, they cannot approach B naively.

3 Statistics at these sample sizes

- **Use a paired test for within-subject data, which is where the design's statistical power actually comes from — and an unpaired test applied to paired data throws the entire advantage away, which is a common and costly mistake.**
- **Check the distribution rather than assuming it. Task completion times are right-skewed rather than normal — a few very slow trials pull the tail — so use a non-parametric test, or log-transform the times and say that you did.**
- **Report the effect size and its confidence interval, not only the p-value — which is CSCE 676 Module 11 §3's argument arriving at the opposite extreme of n: there, large n made everything significant; here, small n makes a real effect non-significant, and the effect size is informative either way.**
- **And count your comparisons. Three metrics across four tasks is twelve statistical tests, which needs a correction (CSCE 676 Module 11 §2) and almost never receives one.**
- **While accepting that with twelve participants you are estimating the effect badly — the interval will be wide — which is a limitation to state rather than to hide (Module 13 §2). The multiplicity problem is as real here as in CSCE 676 and is almost never corrected for: a usability study reporting twelve tests with one at $p < 0.05$ has found exactly what you would expect by chance.**

4 Validity and ethics

- **Demand characteristics. Participants work out what you are hoping for and helpfully provide it — so keep the hypothesis from them, and avoid signalling which of the two designs is yours, which they will otherwise infer from your manner.**
- **The novelty effect. A new design performs well partly because it is new and interesting, and the advantage disappears after a few weeks — which a single session cannot detect at all (Module 06 §4's long-term limit).**
- **Sampling. Your participants are probably people quite like you — colleagues, students, friends — and the population you are designing for is not (CSCE 632's whole argument).**
- **And the practice effect across the session as a whole, which counterbalancing handles within a study and which a single-condition study does not, making before/after comparisons across a session unreliable.**
- **So state your sample's composition — how many, recruited how, with what relevant experience — which is the limitation most often omitted and the easiest to supply. Your participants being people like you is the threat that is hardest to see, because convenience sampling in a university produces a very specific population that resembles nobody's user base.**

AND A STUDY WITH PEOPLE HAS OBLIGATIONS A STUDY WITH DATA DOES NOT

Informed consent: what you will do, what you will record, how it will be stored and for how long, who will see it, and that they may stop at any point without giving a reason and without consequence.

Debrief any deception, including a Wizard of Oz setup (Module 06 §3) — **which is a requirement and not a courtesy**, and which should be scripted in advance rather than improvised at the end of a long session.

Minimise and protect the data. A recording of somebody failing repeatedly at a task is sensitive material, and CSCE 701's handling obligations and CSCE 676 Module 12's privacy reasoning apply to it in full — including the collect-less principle.

And never make the participant feel tested. It is both unkind and methodologically damaging (Module 07 §1) — **which means the ethical point and the practical point coincide here**, and that is a convenient and unusual alignment worth noticing.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Lazar, Feng & Hochheiser — Research Methods in HCI, chapters 2 through 5	The whole module — the designs, the statistics, and the ethics, treated properly. Library copy.
Cairns — Doing Better Statistics in Human-Computer Interaction	§3 specifically — and honest about how badly the field has historically done this. Library copy.
Dragicevic — Fair Statistical Communication in HCI (free)	§3's reporting recommendations — effect sizes and intervals instead of significance, argued for this field.
The Belmont Report, and your institution's ethics guidance (free)	§4's obligations in their original framing — and whatever local process applies to you.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Measure the spread of task times** across ten people on one interface.
- 2 Compare that spread** to the difference you expect between two designs.
- 3 Compute the sample size** a between-subject study would need for that effect.
- 4 Design a counterbalanced within-subject study** for two designs.

- 5 **Identify whether your task has carryover**, and justify the design choice.
- 6 **Plot your task times** and check whether they are normal.
- 7 **Run a paired and an unpaired test** on the same data and compare the p-values.
- 8 **Count the comparisons** in a usability study you have read, and check whether it corrected.
- 9 **Describe your own likely sample** and how it differs from your users.
- 10 **Write a consent script** and a debrief script.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Contrast the two designs on five dimensions.
- 2 Why are individual differences the governing fact?
- 3 Why can within-subject work with a dozen people?
- 4 Name the three order effects and what counterbalancing fixes.
- 5 Which one does it not fix, and what follows?
- 6 Give four statistical rules for small human samples.
- 7 Why does an unpaired test on paired data waste the design?
- 8 Name four threats to validity, and which is hardest to see.
- 9 Give the four ethical obligations.
- 10 Where do the ethical and practical points coincide?